

Multi-Segment Flow Control with Bayesian Occlusion Prediction

Executive Technology Brief

Domain: Hydraulics + Control Systems

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Catheter obstruction causes 30-50% of shunt failures

Mechanism: Multi-segment parallel catheter with Bayesian occlusion prediction and alpha-distribution controller

Architecture: Multi-segment CSF shunt catheter with per-segment flow sensors, Bayesian occlusion predictor, and alpha-distribution controller

Governing Model: Multi-segment CSF shunt catheter with Bayesian occlusion prediction and alpha-distribution controller. Flow redistribution across N segments maintains drainage when one segment occludes.

Key Equations:

$Q = (\pi * r^4 * dP) / (8 * \eta * L)$ [Hagen-Poiseuille per segment]

$G_{total} = \sum(G_i)$ [parallel conductance across segments]

$P(occlude_i | F_{obs})$ updated via Bayesian inference on conductance trend dG_i/dt

$\alpha_i = controller(P(occlude_i))$ [flow redistribution policy]

Kill Condition: Multi-segment shows no advantage over single-segment in $\geq 80\%$ of computational scenarios

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.